

Mining the Relationship Between Music Listening and Mental Health

Jeremy Qiao

CS 4403: Data Mining (Winter 2026)

Instructor: Jake van der Laan

g9rbj@umb.ca

Abstract. The main goal of this research project is to study if there are any relationships between music listening behaviours and mental health conditions. The dataset is made of survey results from 736 respondents and includes information on age, hours of music listened to per day, favourite genres, beats per minute (BPM), and self-reported scores for anxiety, depression, insomnia, and obsessive-compulsive disorder (OCD). The research includes data exploration, data cleaning, correlation analysis, classification, and regression analyses. Overall, while some weak relationships are found between music-related features and mental health, the music features alone do not seem to be strong predictors of mental health scores. In binary classification, accuracy ranged from 0.486 (OCD) to 0.623 (depression). For three-class classification, the best accuracy was 0.45. In regression, the best R^2 was only 0.034. These results suggest that music listening may have some relationship with mental health, but the relationship is limited in this dataset and is not strong enough for reliable prediction.

Keywords: Data Mining · Mental Health · Music Listening · Classification · Regression

1 Introduction

Music is an important part of daily life for many people. People use music for relaxation and emotional support. Because of this, it is interesting to study whether music listening habits are related to mental health. I came to Canada when I was 17. I couldn't speak English at that moment, so music was the only way to make me feel happy instead of using my poor English to talk to other people. So, for this project, I chose to study the relationship between music listening and mental health.

Prior research has studied this relationship from different ways. Gustavson et al. [5] did a broad review of music engagement and mental health, finding positive links between music engagement and lower depression and anxiety symptoms, while also noting that the relationship is complex and that some types of music engagement may be linked to worse mental health outcomes. Wang et al. [6] studied music genre preference and mental health in college students using a self-reported survey, finding that preference for pop and classical music was positively linked to mental health, while heavy music showed a negative relationship. Feneberg et al. [7] used mobile-based daily assessments to study music listening and perceived stress and mood, finding that music listening was linked to lower stress in daily life.

Building on these studies, this project explores the relationship between music listening behaviour and mental health using a publicly available survey dataset [1]. The dataset includes features such as age, hours of music listening, BPM, favourite genre, and whether the person listens while working. It also includes self-reported scores for anxiety, depression, insomnia, and OCD on a 0–10 scale.

The main goal is to find patterns and test whether machine learning models can predict mental health levels from music-related features. Five model types are evaluated: Random Forest, Logistic Regression, SVM, SVR, and Linear Regression. This work was completed as part of the CS 4403 Data Mining course [4].

2 Dataset Description

The dataset was created by Catherine Rasgaitis and is available on Kaggle [1]. It contains 736 survey participants and 33 columns. Before modelling, exploratory data analysis was conducted to understand the structure and distribution of the data.

2.1 Key Variables

Table 1 summarises the main variables used in this project.

Table 1. Key variables in the dataset.

Variable	Type	Description
Age	Numeric	Age of participant
Hours per day	Numeric	Average hours of music listening per day
BPM	Numeric	Preferred beats per minute
While working	Binary (Yes/No)	Whether the person listens while working
Fav genre	Categorical	Favourite music genre
Instrumentalist	Binary (Yes/No)	Whether the person plays an instrument
Composer	Binary (Yes/No)	Whether the person composes music
Exploratory	Binary (Yes/No)	Whether the person explores new music
Foreign languages	Binary (Yes/No)	Whether the person listens in other languages
Anxiety	0–10	Self-reported anxiety score
Depression	0–10	Self-reported depression score
Insomnia	0–10	Self-reported insomnia score
OCD	0–10	Self-reported OCD score
Music effects	Categorical	Improve / No effect / Worsen
Frequency [genre]	Ordinal	How often the person listens to each genre

2.2 Initial Observations

From the first review of the dataset, several important facts were found:

- The dataset has 736 rows and 33 columns. No duplicate rows were found.
- BPM had the most missing values (107 missing, 14.54%). Age had 1 missing value (0.14%).
- One BPM entry was recorded as 999,999,999 and had to be removed as an outlier.
- Age ranges from 10 to 89, with a mean of 25.2 years. Most participants are young adults.
- Mean anxiety score: 5.84; mean depression: 4.80; mean insomnia: 3.74; mean OCD: 2.64.
- 458 out of 736 participants (62.2%) use Spotify as their primary streaming service.
- 579 out of 736 (78.7%) listen to music while working.
- 542 out of 736 (73.6%) report that music improves their mood; only 17 (2.3%) say it worsens it.

Table 2 shows the distribution of favourite genres.

Table 2. Favourite genre distribution among 736 participants.

Genre	Count	Percentage
Rock	188	25.5%
Pop	114	15.5%
Metal	88	12.0%
Classical	53	7.2%
Video game music	44	6.0%
EDM	37	5.0%
Hip hop	35	4.8%
R&B	35	4.8%
Folk	30	4.1%
K-pop	26	3.5%
Country	25	3.4%
Rap	22	3.0%
Jazz	20	2.7%
Lofi	10	1.4%
Gospel	6	0.8%
Latin	3	0.4%

3 Exploratory Data Analysis

I think it is helpful to visualize the data before modeling so that you can see the general shape of the data. At this stage, you can find any distributions with skewness, any outliers or extreme values, and any differences between groups of values that may affect your modeling choice.

3.1 Age Distribution

Most participants are young, mainly between 18 and 25 years old. The highest concentration is around age 20. There are also some older participants, but they are much fewer. The mean age is 25.2 years and the median is 21 years. This age distribution is consistent with the finding by Wang et al. [6] that young adults are the most active music listeners and are also the group most commonly studied in music and mental health research.

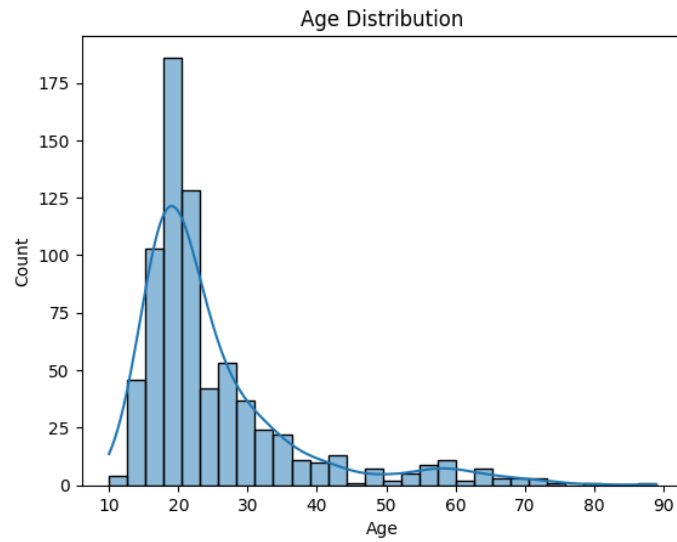


Fig. 1. Age distribution of participants. Most survey participants are young adults, especially around ages 18 to 25. Mean age = 25.2, median = 21.

3.2 Hours Per Day Listening to Music

Most participants listen to music for about 1 to 4 hours per day, with a peak around 2 to 3 hours. The mean is 3.57 hours per day and the median is 3 hours. Very few people listen for extremely long hours, though the maximum recorded is 24 hours.

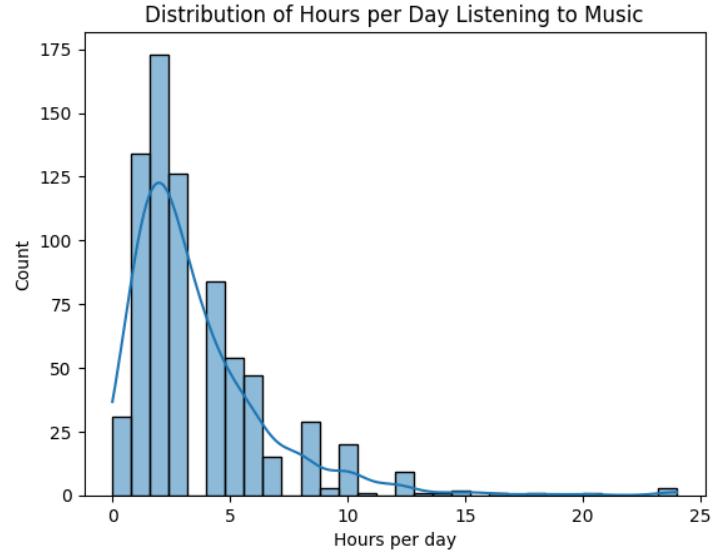


Fig. 2. Distribution of hours per day listening to music. Mean = 3.57 hours, median = 3 hours. Most people listen for a few hours per day, while very high values are rare.

3.3 Instrumentalist and Composer

The dataset also shows whether participants play instruments or compose music. For instrumentalists, 497 participants (67.7%) answered No and 235 (32.0%) answered Yes. For composers, 609 participants (82.9%) answered No and only 126 (17.1%) answered Yes. In both cases, the number of participants answering “No” is much larger than those answering “Yes”.

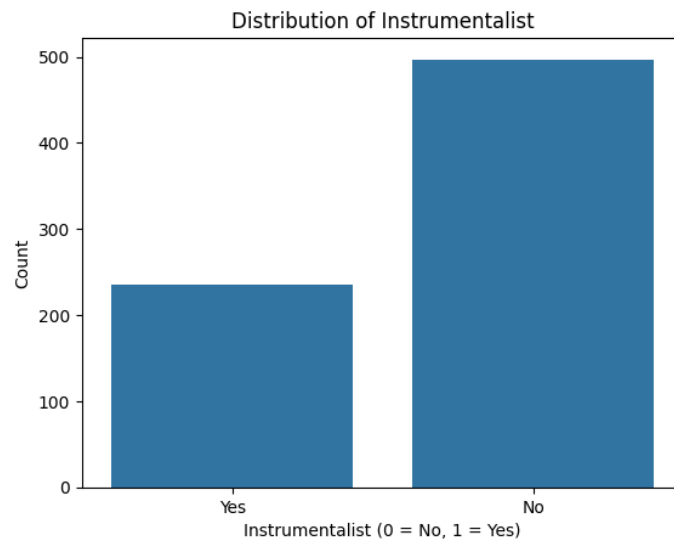


Fig. 3. Distribution of instrumentalist responses. 497 participants (67.7%) are not instrumentalists.

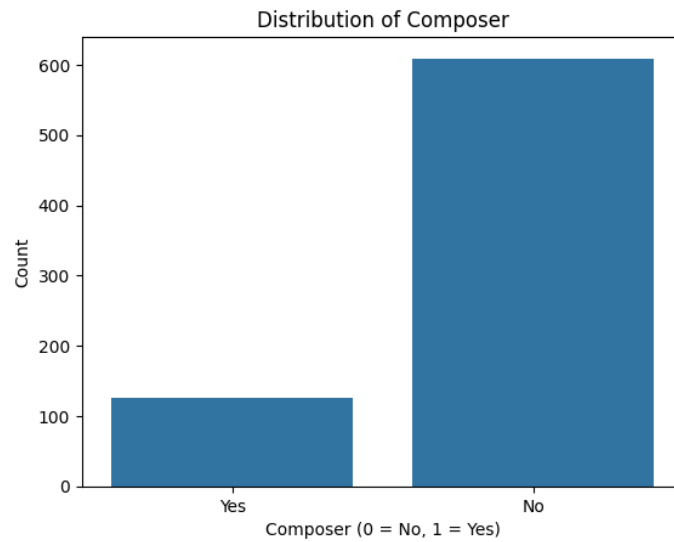


Fig. 4. Distribution of composer responses. 609 participants (82.9%) are not composers.

3.4 Favourite Genre Distribution

Rock is the most popular genre with 188 participants (25.5%), followed by Pop (15.5%) and Metal (12.0%). Together these three genres account for more than half the dataset. Genres such as Lofi, Gospel, and Latin appear much less often.

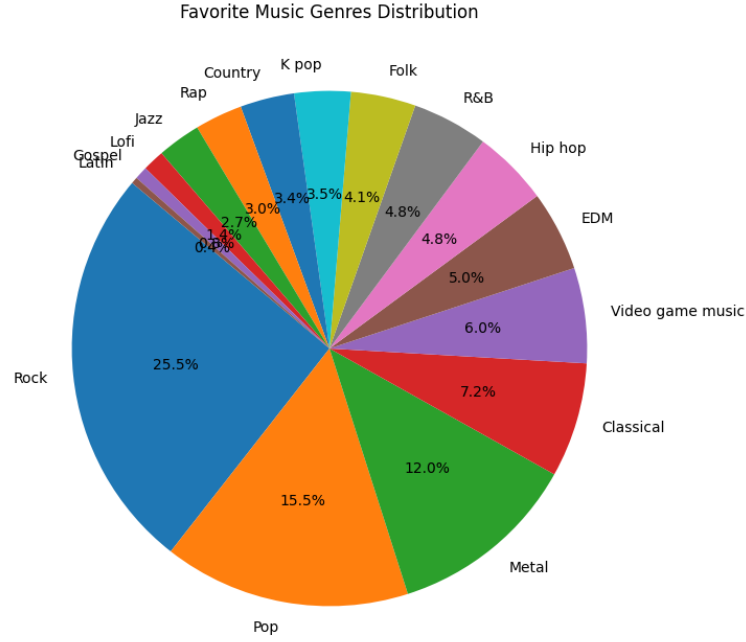


Fig. 5. Favourite music genres distribution. Rock (25.5%), Pop (15.5%), and Metal (12.0%) are the most common genres in the survey.

4 Data Cleaning

Raw survey data usually contains missing values and inconsistent entries. This project cleaned the dataset before modelling.

4.1 Cleaning Steps

The following steps were applied in order:

- Missing Age values (1 row) were filled with the median (21 years).
- Missing BPM values (107 rows, 14.54%) were filled with the median (120 BPM).

- BPM values below 40 or above 250 were removed as unrealistic. This eliminated extreme outliers such as the recorded maximum of 999,999,999.
- Missing Yes/No values in Composer and Instrumentalist were filled with “No”.
- Missing values in Music effects, Fav genre, and Primary streaming service were filled using the most frequent value.
- Four columns not needed for analysis were dropped: Primary streaming service, Permissions, Timestamp, and Fav genre (used for visualisation only, not modelling).

After cleaning, the working dataset had 727 rows (9 rows removed due to BPM outliers).

4.2 Encoding

Since machine learning models require numeric input, text values were converted into numbers:

- Yes/No columns (While working, Instrumentalist, Composer, Exploratory, Foreign languages) were changed to 1/0.
- Frequency columns were mapped to an ordinal scale: Never=0, Rarely=1, Sometimes=2, Often=3, Very frequently=4, Very often=5.
- Music effects were changed into three separate one-hot encoded columns: Music effects_Improve, Music effects_No effect, Music effects_Worsen.

After this step, the dataset had no missing values and was ready for modelling.

5 Correlation Analysis

A correlation heatmap was used to show how variables move together. Frequency variables were removed from this heatmap to make it easier to read.

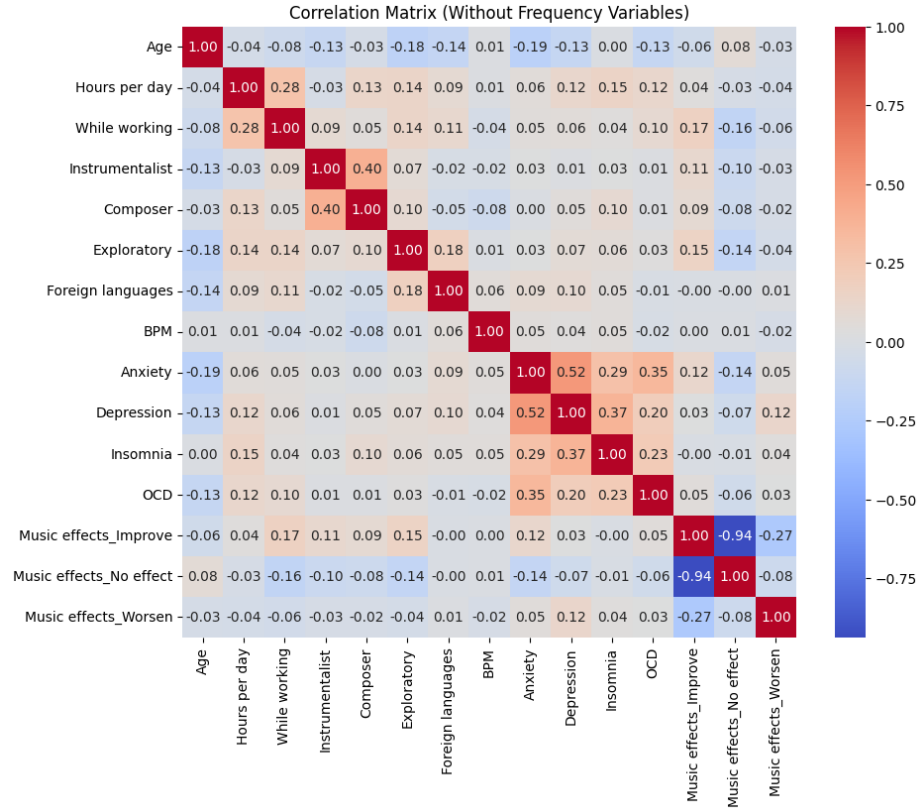


Fig. 6. Correlation matrix without frequency variables. Mental health variables show moderate positive relationships with each other, while music features have mostly weak correlations.

The heatmap reveals several important patterns. Anxiety and depression have the strongest correlation ($r = 0.52$). Insomnia also shows positive correlations with both anxiety ($r = 0.29$) and depression ($r = 0.37$). OCD shows moderate correlations with anxiety ($r = 0.35$) and depression ($r = 0.20$).

In contrast, music-related features show only weak correlations with mental health. Hours per day has a weak positive correlation with overall mental health severity ($r = 0.16$). Among the music frequency variables, Rock ($r = 0.14$), Metal ($r = 0.13$), and Video game music ($r = 0.11$) showed the highest positive correlations with overall severity. Age showed a small negative correlation ($r = -0.16$).

6 Classification Models

This part of the project tested whether music-related features could classify mental health levels.

6.1 Binary Classification

Each mental health variable (Depression, Anxiety, Insomnia, OCD) was turned into a binary problem using the median as the threshold. A Random Forest classifier with 100 trees was trained on each target. Data was split 80/20 for training and testing with stratified sampling.

Table 3 shows the exact results from the notebook output.

Table 3. Binary classification results using Random Forest (Low vs. High, threshold = median).

Target	Accuracy	Precision	Recall	F1 Score
Depression	0.623	0.636	0.709	0.671
Anxiety	0.610	0.648	0.773	0.705
Insomnia	0.589	0.624	0.699	0.659
OCD	0.486	0.518	0.564	0.540

Depression showed the highest accuracy (0.623) and OCD the lowest (0.486). The higher recall than precision across all targets suggests that the models tend to over-predict the “High” class, which means they flag more people as high risk than are actually high risk.

6.2 Three-Level Classification

For a more detailed task, a combined mental health severity score was computed as the average of all four variables:

$$Mental_Severity_All = \frac{Depression + Anxiety + Insomnia + OCD}{4}$$

Participants were then divided into three groups using the 25th percentile ($Q1 = 2.75$) and 75th percentile ($Q3 = 6.00$) as cut-off points:

- Low: score $\leq Q1$ (42 participants in the test set)
- Medium: score between $Q1$ and $Q3$ (64 participants in the test set)
- High: score $\geq Q3$ (40 participants in the test set)

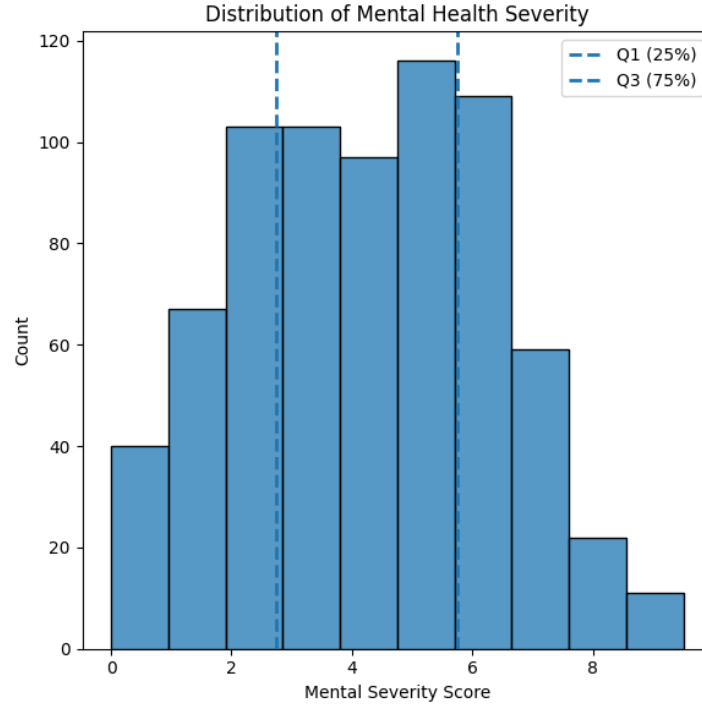


Fig. 7. Distribution of mental health severity. The dashed lines show Q1 (2.75) and Q3 (6.00), which were used to split the data into low, medium, and high groups.

Three classifiers were compared. Table 4 shows the full results.

Table 4. Three-class classification results (Low / Medium / High mental health level).

Model	Accuracy	Macro F1	Medium Recall	Low/High Recall
Logistic Regression	0.390	0.35	0.59	0.26 / 0.20
SVM (RBF)	0.452	0.36	0.80	0.21 / 0.15
Random Forest	0.445	0.40	0.67	0.31 / 0.23

SVM got the highest accuracy (0.452), but it reached this by predicting “Medium” very often (recall = 0.80 for Medium, but only 0.21 for Low and 0.15 for High). Random Forest had the best macro F1 (0.40) and the most balanced performance across all three classes. All models struggled to distinguish the Low and High groups, showing that these groups are not clearly separated by the available music features.

7 Regression Models

Regression models were used to predict the continuous mental health severity score instead of a category. Age was removed from the feature set to keep the analysis focused on only music related behaviour.

Table 5 shows the full comparison of all five models.

Table 5. Regression model comparison for predicting overall mental health severity score.

Model	MAE	RMSE	R^2	Rank
Linear Regression	1.494	1.852	0.034	1st
SVR (RBF kernel)	1.521	1.857	0.030	2nd
Random Forest	1.532	1.872	0.013	3rd
Gradient Boosting	1.592	1.917	-0.035	4th
KNN Regressor	1.534	1.927	-0.045	5th

Linear Regression performed best with $R^2 = 0.034$, second one is SVR ($R^2 = 0.030$). Both Gradient Boosting and KNN achieved negative R^2 values, meaning they performed worse than simply predicting the mean score for every participant. All models showed very low R^2 values overall, that means music-related features alone explain less than 4% of the variation in mental health scores.

To better understand the regression results, these visualizations were created to compare individual features with overall mental health severity.

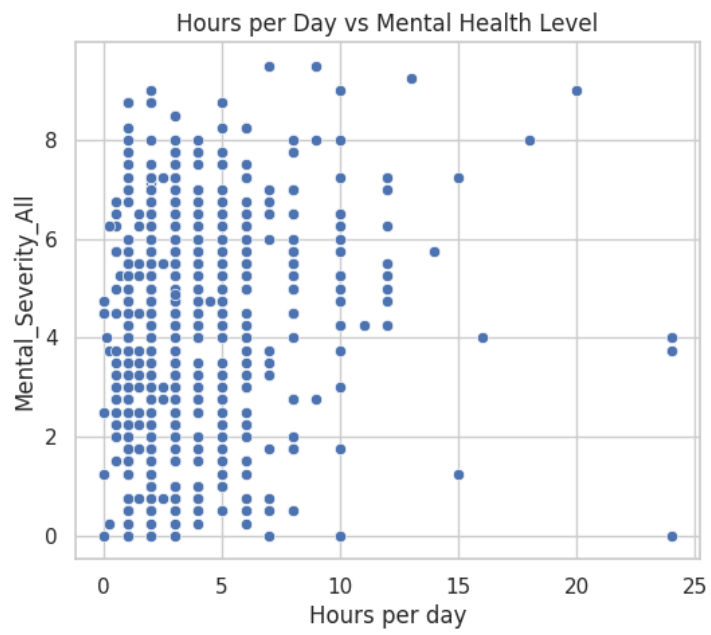


Fig.8. Hours per day versus mental health level. The points are widely scattered, which suggests there is no strong linear relationship ($r = 0.158$).

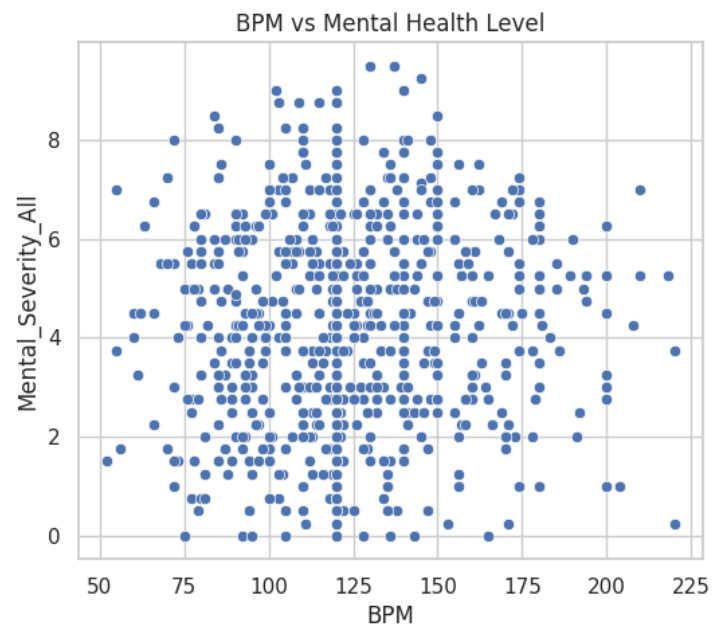


Fig. 9. BPM versus mental health level. The spread of points confirms that BPM alone is not a strong predictor of mental health severity ($r = 0.044$).

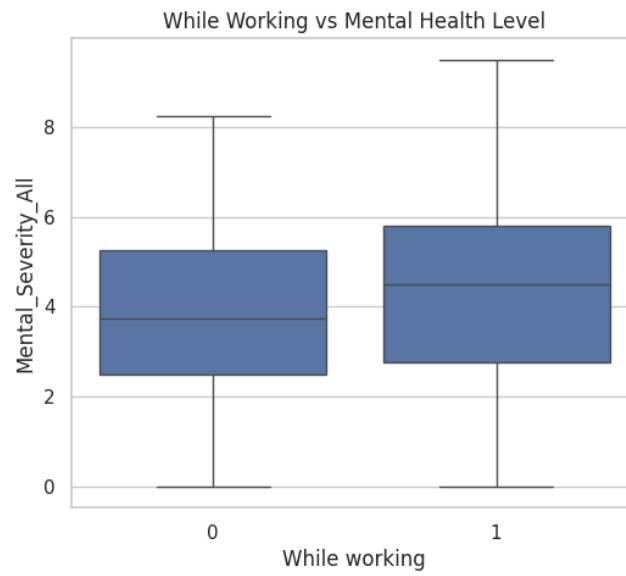


Fig. 10. While working versus mental health level. The two groups show similar distributions, suggesting only a weak difference in mental health between those who listen while working and those who do not.

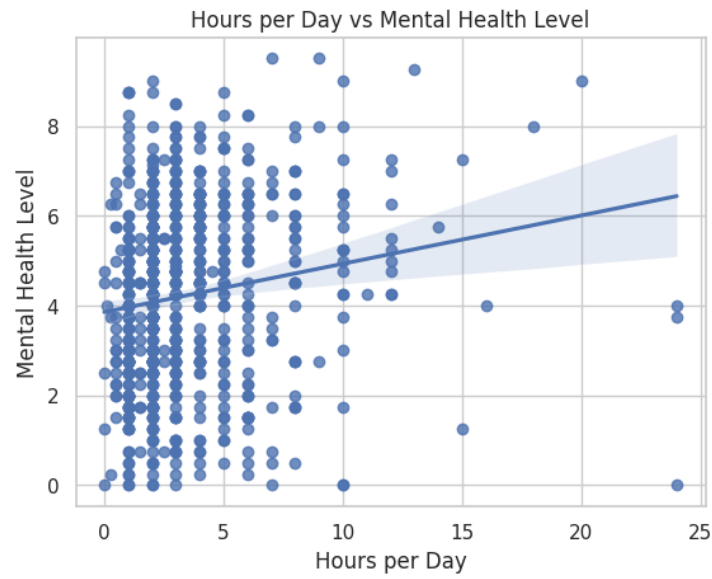


Fig. 11. Hours per day versus mental health level with a fitted regression line. The upward slope is weak ($r = 0.158$), confirming only a small positive trend.

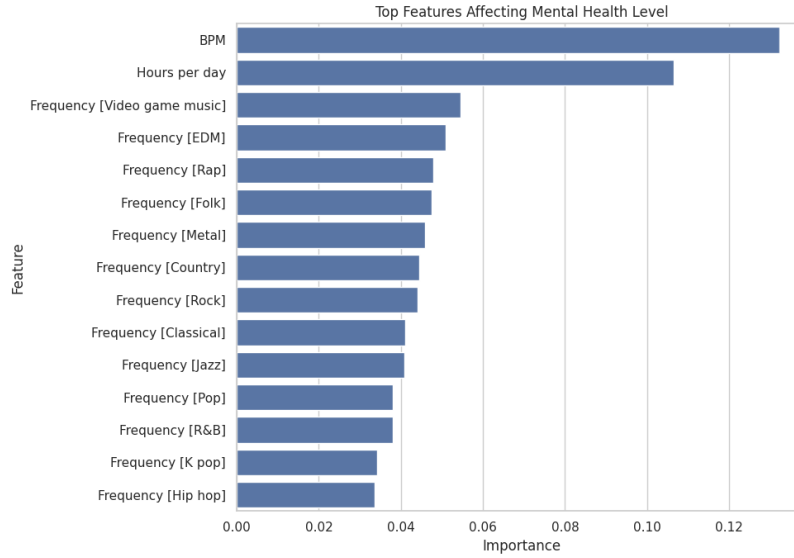
8 Feature Importance and Genre Analysis

8.1 Feature Importance

Random Forest feature importance was used to see which variables contributed the most to prediction. Table 6 shows the top 15 features from the Random Forest Regressor.

Table 6. Top 15 feature importance scores from Random Forest Regressor (music features only, Age excluded).

Feature	Importance Score
BPM	0.132
Hours per day	0.107
Frequency [Video game music]	0.055
Frequency [EDM]	0.051
Frequency [Rap]	0.048
Frequency [Folk]	0.048
Frequency [Metal]	0.046
Frequency [Country]	0.045
Frequency [Rock]	0.044
Frequency [Classical]	0.041
Frequency [Jazz]	0.041
Frequency [Pop]	0.038
Frequency [R&B]	0.038
Frequency [K pop]	0.034
Frequency [Hip hop]	0.034

**Fig. 12.** Top 15 features affecting mental health level (Random Forest). BPM (0.132) and hours per day (0.107) appear as the most important features, followed by several music frequency variables.

To compare results from another model, Gradient Boosting feature importance was used. In Gradient Boosting, BPM scored even higher (0.194) and hours per day also stayed second (0.150), showing these two features are still the useful. But the models still got very low R^2 values, which means high importance does not mean strong prediction performance.

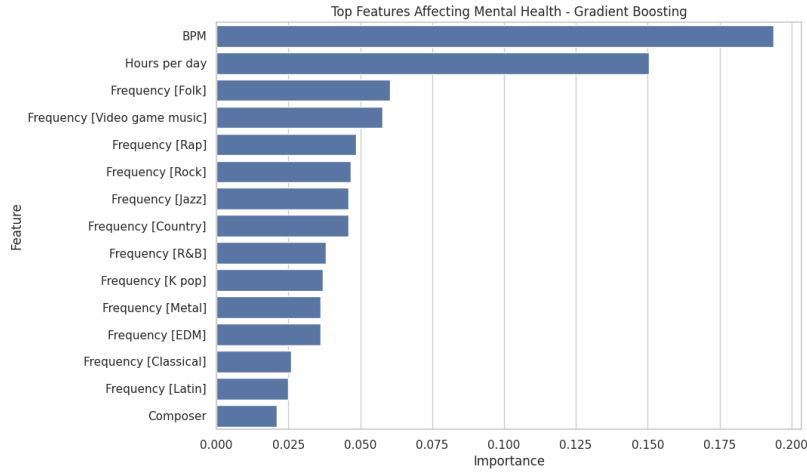


Fig. 13. Top features affecting mental health using Gradient Boosting. BPM (0.194) and hours per day (0.150) remain the strongest features. Frequency [Folk] (0.060) ranked third in this model.

8.2 Genre and Mental Health Scores

I compared average mental health scores through favourite music genres. Table 7 shows the full results for all 16 genres. A similar genre-level analysis was also done by Monfared [3], whose exploratory analysis on the same dataset also found that Lofi and Metal listeners tended to report higher mental health symptom scores.

Table 7. Average mental health scores by favourite genre (all scores on 0–10 scale). Bold values indicate the highest score in each column.

Genre	Insomnia	OCD	Depression	Anxiety
Classical	3.83	2.37	4.15	4.92
Country	2.72	2.76	4.32	5.40
EDM	4.06	3.00	5.19	5.53
Folk	3.69	2.00	5.17	6.52
Gospel	4.40	0.40	1.20	3.80
Hip hop	3.47	2.79	5.76	6.09
Jazz	3.85	2.80	4.50	5.90
K-pop	3.46	2.54	4.42	6.23
Latin	3.33	1.67	3.00	4.33
Lofi	5.60	3.40	6.60	6.10
Metal	4.49	2.33	4.98	5.69
Pop	3.37	2.86	4.49	6.07
R&B	2.97	2.79	3.82	5.18
Rap	2.27	3.18	4.00	5.09
Rock	3.88	2.68	5.24	6.12
Video game music	3.91	2.35	4.40	5.93

The results show clear variation across genres:

- **Lofi** has the highest average scores for insomnia (5.60), OCD (3.40), and depression (6.60). However, the Lofi group contains only 10 participants, so this result should be interpreted with caution.
- **Folk** listeners show the highest average anxiety score (6.52), followed by K-pop (6.23) and Rock (6.12).
- **Gospel** shows the lowest average OCD (0.40) and depression (1.20), though it also has only 6 participants.
- **Rap** listeners report the lowest average insomnia (2.27), while Country listeners report the second lowest (2.72).

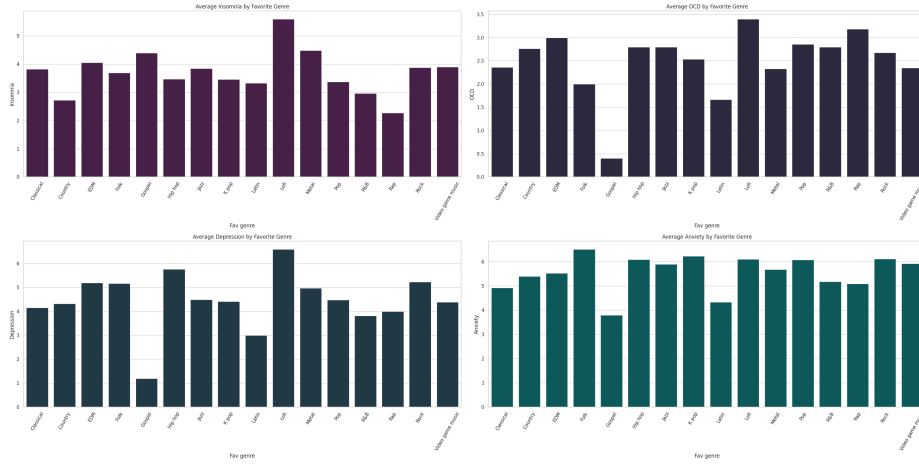


Fig. 14. Average insomnia, OCD, depression, and anxiety by favourite genre. Lofi consistently shows higher average scores. These are descriptive group-level patterns and do not imply that music genres cause mental health conditions.

9 Discussion

The overall results show there are some weak relationships between music listening and mental health, but these relationships are not strong enough for accurate prediction. Correlations between the mental health variables themselves are clearer (e.g., anxiety–depression: $r = 0.52$) than correlations between music features and mental health (highest: hours per day, $r = 0.16$).

This finding is broadly consistent with the review by Gustavson et al. [5], who noted that the relationship between music engagement and mental health is complex and influenced by many factors beyond listening behaviour alone, and that many earlier studies were limited by small samples and methodological constraints. Similarly, Wang et al. [6] found that music genre preference explained only a small portion of mental health variance in their college student sample, which aligns with the low R^2 values observed in this project.

Several possible reasons explain the weak results in this study:

- Mental health is influenced by many factors besides music, such as relationships, work stress, sleep quality, and physical health.
- The data is self-reported, which can introduce noise and reporting bias. Feneberg et al. [7] also acknowledged this limitation in their daily-life music listening study.
- The dataset size (736 participants, 727 after cleaning) is moderate and may be too small to capture subtle patterns reliably.
- Averaging four different mental health conditions into one combined severity score may hide condition-specific patterns. For example, depression and OCD have quite different drivers.

- Some genre groups are very small (Lofi: 10, Gospel: 6, Latin: 3), making their average scores unreliable.

The visualizations also support these conclusions. The scatter plots for hours per day (Fig. 8) and BPM (Fig. 9) show widely scattered points with no obvious trend. The box plot for “while working” (Fig. 10) shows very similar distributions between the two groups. Feature importance plots confirm that BPM and hours per day are the most informative features, but their absolute importance scores are still small (0.132 and 0.107 respectively).

After I talking to Jake, SVM was added as an extra classification model and SVR as an extra regression model. SVM works well for high-dimensional datasets and can model non-linear decision boundaries using the RBF kernel. Since the dataset has many encoded features after one-hot and frequency encoding, SVM and SVR provide a useful comparison with simpler models.

10 Conclusion

This project explored whether music listening habits can predict mental health outcomes using survey data from 736 participants.

My conclusion is: music-related features show only a weak relationship with mental health in this dataset. In binary classification, the best accuracy was 0.623 for Depression. In three-class classification, the best accuracy was 0.452 using SVM. In regression, the best R^2 was only 0.034 using Linear Regression, meaning music features explain less than 4% of the variation in mental health scores. These findings are kind of similar with prior work [5, 6], which similarly found that music listening alone is not a sufficient predictor of mental health outcomes.

Some descriptive patterns were observed, especially in the genre analysis. Lofi fans reported higher average insomnia (5.60), OCD (3.40), and depression (6.60) scores, while Rap and Country fans reported lower insomnia (2.27 and 2.72). These are group-level averages from a survey and do not mean causation.

Future work I could improve on this project in a few ways. A larger dataset would help the models capture more subtle patterns. Including additional features like sleep duration, daily stress levels, or social activity. They could improve predictive power and also study each mental health condition separately rather than combining them into a single severity score (super feature).

My project explains how data mining methods such as classification, regression, feature importance, and correlation analysis can be applied to explore real world behavioural survey data and explain the results in a meaningful way.

References

1. Rasgaitis, C.: Music & Mental Health Survey Dataset. Kaggle (2023). <https://www.kaggle.com/datasets/catherinerasgaitis/mxmh-survey-results>
2. Breiman, L.: Random Forests. *Machine Learning* **45**(1), 5–32 (2001)

3. Monfared, M.: Mental Health & Music Relationship — Analysis & EDA. Kaggle (2023). <https://www.kaggle.com/code/melissamonfared/mental-health-music-relationship-analysis-eda>
4. van der Laan, J.: CS 4403 Data Mining. University of New Brunswick (2025). <https://www.vanderlaan.ca/course-pages/cs-4403-data-mining>
5. Gustavson, D.E., Coleman, P.L., Iversen, J.R., Maes, H.H., Gordon, R.L., Lense, M.D.: Mental health and music engagement: review, framework, and guidelines for future studies. *Translational Psychiatry* **11**, 370 (2021). <https://doi.org/10.1038/s41398-021-01483-8>
6. Wang, K., Gao, S., Huang, J.: Learning about your mental health from your playlist? Investigating the correlation between music preference and mental health of college students. *Frontiers in Psychology* **13**, 824789 (2022). <https://doi.org/10.3389/fpsyg.2022.824789>
7. Feneberg, A.C., Stijovic, A., Forbes, P.A.G., Lamm, C., Piperno, G., Pronizius, E., Silani, G., Nater, U.M.: Perceptions of stress and mood associated with listening to music in daily life during the COVID-19 lockdown. *JAMA Network Open* **6**(1), e2250382 (2023). <https://doi.org/10.1001/jamanetworkopen.2022.50382>